

RBE 4701: AI For Robotics

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Project 1 Report

Approach development timeline

For this project, we had to design an AI to escape a maze in varying difficulties, getting away from different monsters and around walls. Our approach consisted of different solutions and ideas tied together. At first, we decided for a simple A* algorithm with the heuristic being recalculated every step. Walls would receive the highest value for $h(n)$ as they are unreachable. After that, 3 cells around the monsters in the field would receive values of $h(n)$ with the highest being within 1 cell of the monster and the lowest within 3 cells of the monster. Other than that, all cells would have an $h(n)$ of 0. This performed well for variants 1-3, but when it came to deal with variants 4 and 5, it struggled. The character would move in a safe direction, but then the monsters would move towards the character at the same time, detect it, and then proceed to eat it. We then tried an approach in which the character would get away from the monster or continue an optimal path (recalculated whenever we escape from a monster) based on how close the monster is and how far the exit is. We called this how “brave” the character should be. Bravery proved to be effective for variant 5, but for variant 4 it would struggle to reach the 5/10 mark to pass. Finally, we settled on the actual

approach we used, built around always trying to stay out of sight from the monsters based on their movements.

The Chosen Approach: Safe Distance

After realizing we could see the direction the monster will move in, we realized staying a safe distance away is easier than we originally thought (Note we only look at the closest monster because it's the most dangerous one). Simply put, if the monster never detects you then it won't eat you. Therefore, we determine if a move is safe depending on if, after the monster and the character move, the x or y distance between them is 3 cells (as the highest sighting range of the monsters is 2 in variant 4).

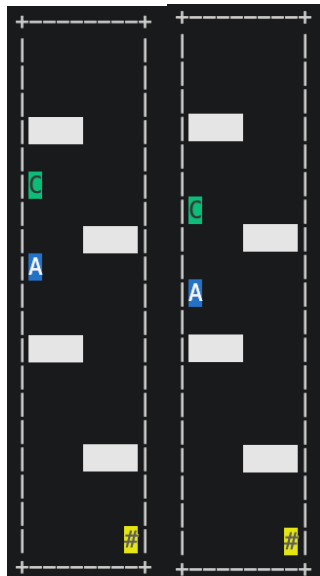


Figure 1: Safely following a monster

In Figure 1 above we can see the hero progress towards the goal moving towards a monster while maintaining a safe distance of 3. This way, we can progress as much as possible without getting detected using the monster's movements against it. If the distance isn't safe,

we move in the opposite direction from the monster to get away from it. Of course, getting away is sometimes not possible and our hero gets cornered or backed against the wall.

Getting Cornered

Getting cornered often happened when the monster headed for the hero diagonally or straight up above the bottom left wall. When the monster is heading straight for the hero and the hero doesn't try to change direction, it's doomed. We therefore added a check which said, if the monster's x direction isn't changing and neither is the hero's when backed into a wall on the y direction, move 1 in the x direction to put some distance between the hero and the monster (and vice versa if stuck in the y). This way, the hero gets a bit farther from the monster, and now the monster isn't straight ahead and the hero can move the opposite direction from it.

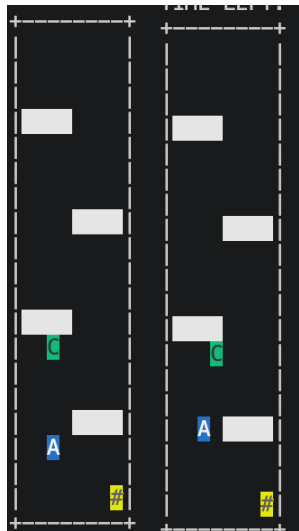


Figure 2: A successful wall escape

Figure 2 above shows how the hero gets away when backed up, giving it the chance to keep running away and live longer, hopefully losing the monster in the process.

Running for it

Even with these 2, something was missing. The hero would choose to run away when victory was in sight instead of going for it. To fix this, the hero goes straight for the exit and is very brave when it's below the monster. Even if the monster sees it, the distance between them should be at least 3 tiles, so the chances it's caught before reaching the exit when moving on the optimal path (which is guaranteed from A*) are very low. Figure 3 below shows one such case where unsafe moves were made (dx towards the monster instead of away) because the hero was below the monster running for the exit.

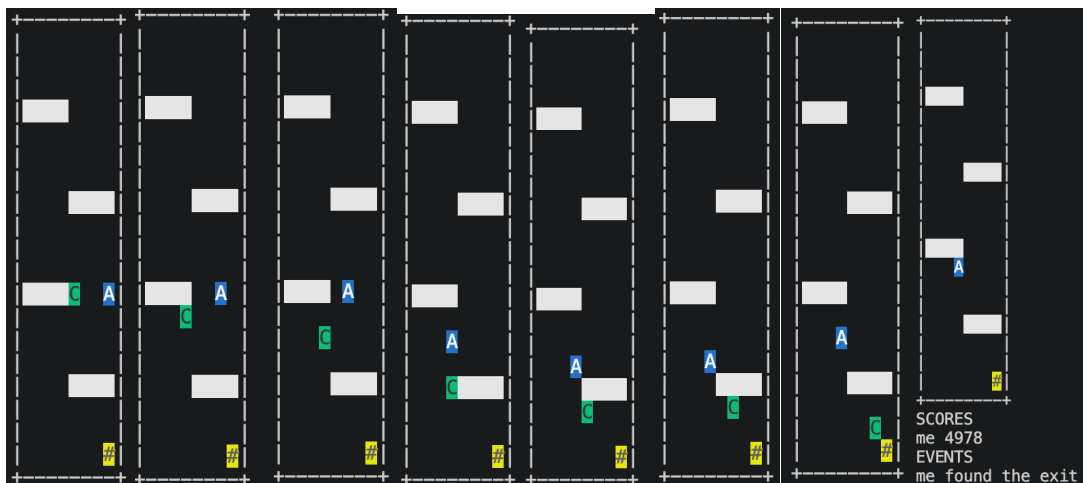


Figure 3: Running for it even if it's not the safest

With these 3 aspects, our hero can clear all variations within the expected times.

Evaluating

We ran each variation 30 times to get an idea of how well it performs after a long period of time, with different seeds each time.

Variant	Exits	Deaths	Success Rate
1	30	0	100%
2	25	5	83.3%
3	26	4	86.7%
4	18	12	60%
5	17	13	56.7%

Getting cornered for variants 2, 3, and 4 (more for 4 of course) is sometimes inevitable if the monster moves diagonally toward the hero with a wall behind, leading to a higher number of deaths. Similarly, in variant 5 getting sandwiched between two monsters and cornered can be inevitable. As such, the deaths can be sometimes higher than the exits on small samples of variant 5. However, after a large enough amount of runs there should be slightly more exits than deaths in variant 5.